

# Satyam Awasthi

Researcher & Software Engineer

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## Professional Summary

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Computer Science researcher specializing in **Autonomous Decision-Making**, **Spatial AI**, and **Embodied AI** within **Cyber-Physical Systems**, with a core focus on engineering verifiably safe, constraint-bound execution loops.

## Education

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**University of California, Santa Barbara (UCSB)**

*M.S. in Computer Science (GPA: 3.91/4.00)*

Graduate Research focused on Spatial AI, AR/VR, and HCI at the Four Eyes Lab.

Advisor: [Prof. Tobias Höllerer](#) · Co-Advisor: [Prof. Michael Beyeler](#)

Santa Barbara, CA

Sep 2021 – Jun 2023

**Indian Institute of Technology, Kharagpur (IIT Kharagpur)**

*B.Tech. in Electrical Engineering, Minor in Computer Science (GPA: 8.98/10.0)*

Graduated with First Class Honors.

Kharagpur, India

Jul 2016 – Jul 2020

## Skills

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**Programming Languages:** Python, C++, Kotlin, Java, MATLAB

**Frameworks & Core Tools:** Android/iOS Native SDKs, LangChain, Pydantic (JSON Schema), OpenCV, ROS (Robot Operating System), PyTorch

**Fields of Interest:** Autonomous Decision-Making, Embodied AI, Spatial AI, Multi-Modal Sensor Fusion, Cyber-Physical Systems (CPS)

## Publications

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### Refereed Conference Papers

Snigdha Das, Satyam Awasthi, Abdul Shamnar P, Pradipta De, Sandip Chakraborty, and Bivas Mitra.

**SmartWatch for Wall Writing: Real-time Transcription of Wall Writing from Inertial Sensing**

*Proceedings of the 14th International Conference on COMmunication Systems & NETworkS (COMSNETS)*, 2022.

DOI: [10.1109/COMSNETS53615.2022.9668531](https://doi.org/10.1109/COMSNETS53615.2022.9668531)

Created inertial tracking methods to decode architectural hand stroke motions into alphanumeric symbols on wrist-worn wearables.

### Refereed Poster Papers

Satyam Awasthi, Vivian Ross, Sydney Lim, Michael Beyeler, and Tobias Höllerer.

**Eye Tracking Performance in Mobile Mixed Reality**

*IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, 2024.

DOI: [10.1109/VRW62533.2024.00321](https://doi.org/10.1109/VRW62533.2024.00321)

Investigated low-latency mobile eye tracking performance profiles across modern mixed reality layouts. Supported by NSF Grant #2211784.

Satyam Awasthi, Vivian Ross, Michael Beyeler, and Tobias Höllerer.

**EyeTTS: Evaluating and Calibrating Eye Tracking for Mixed-Reality Locomotion**

*IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)*, 2023.

DOI: [10.1109/ISMAR-Adjunct60411.2023.00104](https://doi.org/10.1109/ISMAR-Adjunct60411.2023.00104)

Proposed unique calibration logic pipelines specifically optimized for eye tracking modalities within mixed-reality locomotion patterns. Supported by NSF Grant #2211784.

## Experience

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### Intuit Inc

Mountain View, CA · Mar 2024 – Jan 2026

*Software Engineer 2*

- Built **dual-stage agentic decision pipelines** using tool-schema routing for real-time intent classification and sequential state tracking.
- Developed a decoupled **generative UI engine** leveraging RAG and structured LLMs to optimize deterministic component layouts.

### Yahoo Inc

Mountain View, CA · Apr 2023 – Nov 2023

*Software Dev Engineer II (IC3)*

- Developed **client-side rule engines** for asynchronous receipt payloads, executing deterministic filtering for **real-time decision support**.
- Designed a dynamic, state-driven carousel UI framework optimized for client-side rendering efficiency.

### Coupang Global LLC

Mountain View, CA · Jun 2022 – Sep 2022

*Software Engineering Intern*

- Engineered **context-aware heuristics** to prune multi-dimensional search spaces based on active user state vectors.
- Designed **Content-Based Filtering (CBF)** layers using **cosine similarity** to rank high-dimensional product features.

### Samsung Research Institute

Noida, India · Sep 2020 – Sep 2021

*Software Engineer (CL 2)*

- Architected a **continuous on-device perception pipeline** ingesting multi-modal media streams for automated ML classification.
- Designed a **cross-profile data brokering engine** leveraging multi-user storage isolation to route assets from **User 1** to sandboxed **User 0**.

## Projects

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### JitterNot

Jan 2022 – Mar 2022

*LSTM-Based Model Predictive Controller for Adaptive Video Streaming*

UCSB

- Developed an LSTM network for real-time throughput prediction, transforming the closed-loop control system from MISO to SISO.
- Implemented an MPC controller leveraging Receding Horizon Control (RHC) principles to dynamically calculate optimal video resolution.

### Autonomous Ground Vehicle & Swarm Robotics

Jan 2016 – Dec 2018

*End-to-End Perception, Planning, and Multi-Agent Navigation Stack*

IIT Kharagpur

- Built an autonomous navigation stack in ROS, fusing heterogeneous sensor streams (LiDAR, IMU, GPS) via an Extended Kalman Filter (EKF).
- Implemented visual obstacle segmentation networks (U-Net) along with global ( $A^*$ /Dijkstra) and local (DWA) path planners.
- Researched decentralized swarm coordination using peer-to-peer relative pose estimation (AprilTags) over an ad-hoc XBee mesh network.

### In Motion — HAR using DeepConvLSTM

Nov 2019 – Nov 2019

*Distributed Mobile Computing System for Human Activity Recognition*

IIT Kharagpur

- Designed a client-server telemetry pipeline streaming high-frequency IMU sensor arrays for rolling window inference.
- Implemented a DeepConvLSTM core architecture using TimeDistributed CNN layers to model spatial-temporal features efficiently.

### CohesiveAR

Apr 2022 – Jun 2022

*Augmented Reality Pipeline for Texture Extraction and Homography Mapping*

UCSB

- Engineered an asset ingestion pipeline using Google ARCore and OpenCV to project 3D world coordinates into 2D display spaces.
- Implemented perspective warp and homography transformations to rectify camera tilt skew into standardized orthogonal textures.

### Tweeter: Distributed Infrastructure Testbed

Sep 2021 – Dec 2021

*High-Concurrency Microblogging Testbed on AWS EC2 Clusters*

UCSB

- Conducted high-throughput load testing up to 512 users/sec, mapping user trajectories as state machines to latency scaling curves.

- Optimized read throughput by deploying nested fragment caches and database query preloading to eliminate bottleneck points.

## Teaching

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**University of California, Santa Barbara**

*Graduate Teaching Assistant*

Santa Barbara, CA

Sep 2021 – Jun 2023

- Directed instructional labs, formulated algorithmic programming assignments, and monitored system architecture milestones for senior capstone teams.
- Covered advanced and core tracks: CS189 (Capstone), CS184 (Android), CS24 (C++ Data Structures), and CS16 (Problem Solving).

## Academic Service & Peer Review

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**Technical Program Committee & Peer Reviewer**

*ACM Symposium on User Interface Software and Technology (UIST)*

May 2024

Served as an external peer reviewer evaluating full paper submissions for novel interface hardware, interactive system architectures, and spatial tracking methodologies.